

SceneForge Outliner

A scene browser that does not freeze Max. This guide describes the release for 3ds Max 2027 and gives numbers measured on a scene of 50 026 objects.

panel v135 .NET 10 core Qt 6.8.3 / PySide6 reference scene SF-50k [Wersja polska](#)

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01 Installation

Drag the received `.mzp` onto a 3ds Max viewport. The installer verifies every copied file by size and finishes with a check that walks the disk from scratch — if anything is missing it names it, instead of writing “done”.

This release requires 3ds Max 2027. The installer checks the Max version and stops if it does not match — instead of leaving you a full set of files and a panel that will not open.

The three places the files go:

WHAT	WHERE
core, panel, columns, scene listener	scripts\SceneForge\
the commands visible in Customize	usermacros\SceneForge.mcr
registration of the menu entry	scripts\startup\

Installing is repeatable: on every update the older panel versions are **removed** once the new ones are safely in place, and the final check requires exactly one panel to be left in the folder.

Uninstalling

```
(SF_odinst = getFiles ((getDir #userScripts) + "\\SceneForge\\SceneForge_odinstaluj_v*
```

It removes the columns from the native explorer, unregisters the scene listener, releases the core and deletes the files. **It does not touch** your keyboard assignments or panel preferences — after reinstalling, everything is as it was.

It removes only the files it put there itself — the installer leaves a list of them in the plugin folder. Your own script or note kept in that folder stays untouched, and so does the folder itself; the uninstaller writes down what it left behind and why.

02 First run

The panel does not start together with Max. Loading the core costs about a second and is meant to happen when you need it — not on every launch of the program.

1. **Customize** → **Customize User Interface**, category `SceneForge`.
2. There are two commands there: *SceneForge Outliner* and *SceneForge - Scene reports*. The command names stay in English in both language versions — Max remembers your keyboard assignments by them, so translating them would lose your settings. If you also installed the free *Explorer Turbo*, its command sits in the same category — that is the third one.

3. Assign a key to them, put them on a toolbar or into a quad menu.

The entry in the main Max menu appears **after restarting** the program. That is not an oversight: in Max 2025+ the menu manager arrives only as a notification parameter, at the moment Max rebuilds the menus itself.

679 ms first run of the command	0–1 ms later ones — the panel only shows itself	101 ms snapshot of a 50 026 object scene
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The Menu button

The first button on the bar: **Menu** — every panel command in groups (List and view, Viewport, Layers, Selection, Material, Objects, Reports and rules, Settings and tools), with tooltips and shortcuts on the right. The same as the palette (Ctrl+P), but without typing — the palette only works while the panel has focus, because in Max Ctrl+P is Pan. Next to Health there is a **Reports** button, and the side bar carries the Reports and Rules icons.

What is new in v121–v133

WHERE	WHAT	HOW IT WORKS
Reports > Timeline	Names of deleted objects, Ctrl+Z as its own kind	The name is recorded at the moment of the event, and deleted objects get their name from the snapshot difference (“Box001 (deleted)”). Undo and redo are recognisable in the table.
panel	Selection history: Alt+Left / Alt+Right	Up to 50 recent selection sets, like back and forward in a browser; a new selection made after going back cuts off the “forward” branch. Objects that no longer exist are counted.
above the list	View tabs	Saved views as tabs, in the order from the Manage views window (the same one as Ctrl+1...9). A click loads the view.

WHERE	WHAT	HOW IT WORKS
palette	Annotations: export / import JSON	Tags, notes and pins by object name, out to a file and back (another version of the file, a colleague). The status bar lists the names that are missing; the import is one undo block.
palette	Merge a file with name checking...	Names read from a .max file without opening it, a list of collisions with the scene, a prefix for the colliding ones; merged objects keep their names instead of Max's Box002. Note: the merge itself is not undone by Ctrl+Z (mergeMAXFile does not record history); the merged objects are left selected.
palette	Differences between two .max files...	Only in A, only in B, common to both, by name and without opening the files; CSV. A rename cannot be recognised by name.
Preferences › Render queue	EXPERIMENTAL: Render the selected cameras one after another	Each camera with the current renderer into folder\camera_name.png at the current resolution. Measured on Scanline, Arnold, V-Ray 7, Corona and FStorm. Max is blocked for the duration of the queue; the time depends on the renderer's own limits.
Preferences › Język / Language	English interface (the default) or Polish	Labels, tooltips, menus, messages and the filter cheat sheet in the chosen language; changing it reloads the panel. Your own shortcuts and macros work in both languages (they are compared by key). The documentation comes in two versions: English (the default) and Polish. The installer copies both, side by side.
side bar	Presentation mode icon	The same as Ctrl+Alt+X; the icon state follows the mode.
chip bar	All / None / Invert	Like Display: All / None / Invert in the Scene Explorer; they act on the category chips, with a single filter recalculation.
Preferences › Expert mode	EXPERIMENTAL: the panel stays in Expert mode	Ctrl+X in Max hides every dock; with this option the panel asks about Expert mode every 500 ms and returns to its place once the mode is on. Not verified with a floating panel or across several monitors.

WHERE	WHAT	HOW IT WORKS
Ctrl+Alt+X	Presentation mode	Only the filter field and the list remain, font +3 pt; pressing it again restores exactly the previous layout. For video and for a laptop.

What is new in v106–v118

WHERE	WHAT	HOW IT WORKS
dock	Panel width	The toolbars wrap their rows, so a docked panel can be made narrow (the minimum dropped from 1218 px to about 240 px).
row menu / palette	Paste names from the clipboard and select	The reverse of “Copy names”: names one per line, separated by commas or semicolons, become a selection; the status bar lists the ones that are missing.
layer list	Solo a layer (Alt+click)	Only that layer stays on, the rest go off with their layout remembered; a second Alt+click restores it. One undo block. Also in the layer menu and the palette.
list	Cameras and lights	A double click on a camera points the viewport at it; row menu: “Look through this camera”, “Turn lights on / off” (on or enabled, one undo block).
above the list	Path bar	The ancestors of the current object as buttons (by handle, not by name); a click selects the ancestor. Palette: “Path bar: show or hide”.
columns	Age	Time since the object last changed in this session (12 s, 5 min, 2.3 h); empty = untouched. The same counter as the filter changed: .
columns, row menu, filter	Tags and notes	User properties <code>sf_tag</code> (labels separated by commas, a coloured

WHERE	WHAT	HOW IT WORKS
		dot) and <code>sf_note</code> ; they are saved with the .max file and are visible in Object Properties > User Defined. Filter: <code>tag:urgent</code> , <code>tag:*</code> , <code>tag:0</code> , <code>note:word</code> .
filter	<code>map:</code>	Objects whose material uses a map whose file matches the pattern: <code>map:wood*</code> , <code>map:*.hdr</code> , <code>map:tiles</code> (a substring of the file name).
lens by material	Drag rows onto a material header	The objects take the material of that bucket (onto “(no material)” – the material is removed). One undo block.
palette	Object visibility states...	Named snapshots of hidden and frozen flags per object, stored in the scene file; restoring one sets exactly that layout, undoable with a single Ctrl+Z.
Preferences > Scene file	Panel settings inside the scene file	When a .max is saved the panel adds the filter, lens, columns, pins and budgets; opening a file that carries them brings them back. On by default; the palette also has “save now”.
palette	Macros	“Macro: start recording” notes filters, palette/Menu/shortcut commands and row menu actions; “stop and save...” gives it a name. Every macro is a palette command (“Macro: name”), so it can get its own shortcut. The “Macros...” window runs and deletes them.
Preferences > Footer counters	EXPERIMENTAL: faces of the selection and of the scene	Off by default. The face total for the whole scene requires counting the faces of every object – on a large scene the first count takes seconds and freezes Max. Turning it on shows a warning.

WHERE	WHAT	HOW IT WORKS
Delete a layer together with its sublayers	Ctrl+Z undoes all of it	Measured end to end: one undo brings back the layers, the sublayers, the objects and the assignments. The old warning “do not rely on Ctrl+Z” was a measurement error and is gone.
Batch rename	Regular expressions	These were there already: the “find” field with “regular expression” ticked, and capture groups in “replace” (\$1).

Undo and redo (v105)

Every panel operation that changes the scene is one entry in the 3ds Max undo history: rename (F2, batch, template), hide, freeze, renderability, display properties, colour, material, an object's layer, hierarchy, hiding and freezing a layer, a new layer, renaming a layer, nesting layers, selection sets, merging duplicates, scene rules, Scene Doctor, viewport painting. Measured by a probe on each of these operations: operation → *max undo* → the scene state returns to the earlier one (30 out of 30), and *max redo* brings the change back.

WHERE	WHAT	HOW IT WORKS
panel, focus in the list	Ctrl+Z / Ctrl+Y / Ctrl+Shift+Z	Max does not receive these keys from a docked window, so the panel intercepts them itself and runs <i>max undo</i> / <i>max redo</i> . In the filter field and in the name editor Ctrl+Z undoes typing, as everywhere else.
Menu > palette	Undo the last operation, Redo	The same commands from the palette (Ctrl+P) and from the Menu button.
Ctrl+Z in the viewport	the list refreshes by itself	After an undo or redo (from the viewport as well) the panel synchronises its snapshot with the scene.
outside the history	isolation, viewport filter	These are modes, not edits: you leave them with a button, not with Ctrl+Z (exactly like the native isolation).

Production tools (v095–v104)

WHERE	WHAT	HOW IT WORKS
Reports > Textures	Relink the missing ones...	You point at a folder (subfolders included); every map whose file is missing from disk gets the first file with the same name. BitmapTexture, V-RayHDRI, CoronaBitmap, Arnold and any map with a <code>filename</code> . One undo block; a summary of what is still missing.
Scene health	Fix	Unambiguous fixes with one click: zero or negative scale → 1,1,1; far from the origin → 0,0,0; no material → <i>SceneForge_Domyslny</i> ; duplicate names → a number; hidden and frozen → revealed; empty layers → deleted; meshes with no faces → deleted. A confirmation with a count, one undo block.
row menu	Name from a template...	<code>{layer}_{class}_{##}</code> , <code>{mat}</code> , <code>{name}</code> ; a preview of old → new before it is applied.
Menu > Layers	Export layers to files...	Every layer that has objects into its own <code>.max</code> (saveNodes) or <code>.fbx</code> (the current FBX settings), named after the layer.
row menu > Material	Replace this material across the whole scene with...	The material of the selected object is replaced by one chosen from the scene, on every object that uses it; the list shows usage counts.
row menu	Select objects similar to this one	The same class, faces and size $\pm 20\%$, the same material — the query lands in the filter field (so you can adjust it) and the result is selected.
right click on a layer	Face budget for the layer...	The layer list shows <i>faces / budget</i> and turns red once it is exceeded. 0 removes the budget.
row menu	Pin (the pin above the list)	A bar of pinned objects above the list, independent of the filter; a click selects and scrolls to them, a right click unpins.
Preferences	Automatic rules	Opt-in: after a file is opened and after an import the panel waits 1.5 s, rebuilds the list and runs every scene rule.
side bar	X-ray the rest	The selection stays solid, everything else becomes See-Through; a second click removes it from exactly the objects it was given to.

Context menu

Right click on a row. The entries the native explorer does not have:

SUBMENU	ENTRY	WHAT IT DOES
Material	Open the material in the editor	In Slate mode it creates or finds the material node in the active view, selects it and puts it into the parameter editor; in classic mode it puts the material into the active slot and opens the editor.
Material	Assign a material from a slot	A submenu of the 24 classic editor slots, read at the moment it is opened. It assigns to everything selected.
Material	Assign the material from the Slate editor	The material Slate holds in its parameter editor goes onto the selection.
Material	Copy from the first selected object	The material of the topmost selected row onto the others — the same instance, not a copy.
Material	Remove the material	It takes the material off; the material itself stays in the scene if something else uses it.
Layer	New empty layer...	It creates an empty layer and makes it current (like <i>Create New Layer</i> in Max). The plus button on the side bar does the same; the layer menu additionally has <i>New layer UNDER...</i> , which nests it straight away.
Selection	Add to a set / Remove from a set	It adds the selection to an existing Max selection set, or takes it out. The list of sets is read when the menu opens — a set created a moment ago in the Max toolbar is already there. A set that has become empty is deleted.

After every operation that changes how the scene looks (hide, freeze, *As a box*, *See-Through*, layers, isolation) the core **redraws the viewport immediately** — previously the change only showed up once you moved the view.

Each of these operations goes into a single undo block — `Ctrl+Z` undoes all of it. After a material change only the **rows that were touched** are refreshed, so the filter and the lens stay where they were.

03 Filter language

The filter field accepts a plain substring, but it also understands a syntax that replaces the *Advanced Filter* window. Conditions are joined with a space (a conjunction), and each of them can be negated with a minus: `-hidden:1`. Parentheses with a vertical bar mean “or”: `(layer:A | layer:B)`.

```
type:VRayLight layer:02_* poly:>100k -hidden:1
```

THE FIELD SUGGESTS AS YOU TYPE

You do not have to know the tables below by heart. Start typing and you will see the keys; **after the colon — the values from this scene**, with object counts, so you can see straight away how many objects you will get. A list of classes that are not in the scene would be noise, so we do not show it.

Only the **term under the cursor** is completed, so the rest of the query stays untouched. A value containing a space is quoted automatically. When there are more matches than the list can hold, the last entry says how many are left — a missing entry never pretends to be a missing layer.

Prefixes

SYNTAX	MEANS
type: class	the object class, not localised — <code>Box</code> , <code>Editable_Poly</code> . Also <code>klasa:</code>
layer: name	the layer; an asterisk works as a wildcard. Also <code>warstwa:</code>
mat: name	the material. Also <code>material:</code>
poly:>100k	the face count; operators <code>></code> <code><</code> <code>>=</code> <code><=</code> . Also <code>sciany:</code>
depth:>2	the depth in the hierarchy. Also <code>glebokosc:</code>
children:>0	the number of children. Also <code>dzieci:</code>
instances:>1	the number of objects sharing the mesh. Also <code>instancje:</code>
selset: name	a named Max selection set. Also <code>zestaw:</code>
~word	a fuzzy search in the name: one typo, a missing letter or one letter too many still matches (<code>~teapot</code> finds <code>Teapto01</code>). Words shorter than 4 characters are not fuzzed.
changed:<10m	objects changed within a time window, according to the event listener: <code><10m</code> , <code>>2h</code> , <code><45s</code> , <code>>1d</code> . The chip <i>Last 10 min</i> does the same; <code>is:recent</code> = 10 minutes.
is:changed	added or changed since the remembered baseline (Reports → Snapshot comparison).
size:>1000	the largest bounding box dimension in world space (with the node scale), in scene units; geometry only — lights and cameras drop out of every comparison. The first use

SYNTAX

MEANS

computes the bounding boxes of all objects (the same cost as `poly:`), after that it comes from memory. Also `rozmiar:`

cat: category the category by the SuperClassID of the base object: `Geometry` , `Shape` , `Light` , `Camera` , `Helper` , `SpaceWarp` , `System` — the same thing the native explorer filters by. `cat:Light` finds the lights of every renderer without knowing the class names. A typo gives an error with the list of categories, not an empty list. Also `kategoria:`

group: name the group. Also `grupa:`

hidden:1 hidden. Also `ukryty:`

frozen:1 frozen. Also `zamrozonny:`

is: trait a scene health chip — see below. Also `jest:`

Chips `is:`

CHIP

FINDS

is:zeroscale objects with zero scale — invisible in the render, yet still in the scene

is:nonuniform non-uniform scale, a common cause of trouble on export

is:faraway objects far away from the rest of the scene; usually a mistake while pasting

is:missingmat with no material. Also `is:bezmateriálu`

is:orphan with no parent and no children. Also `is:sierota`

is:emptylayer empty layers

is:unresolvedxref unresolved external references

is:nolayer with no layer. Also `is:bezwarstwy`

is:leaf leaves of the hierarchy. Also `is:lisc`

is:root roots. Also `is:korzen`

is:unique a mesh used only once. Also `is:unikat`

is:instance an object sharing a mesh. Also `is:instancja`

is:reference a reference. Also `is:referencja`

CHIP	FINDS
is:selected	selected in the scene. Also <code>is:zaznaczony</code>
is:grouphead	a group head
is:groupmember	an object inside a group

Query history

The *hist.* button next to the field drops down the recent queries (up to 20, newest on top); a click puts the query into the field. Only queries the core accepted without an error get there — an unclosed parenthesis typed halfway does not. The history is remembered between sessions. The command *Select by size...* in the palette asks for a condition (e.g. `>1000`), writes `size:` into the filter and selects everything that is left.

Category and state chips

Below the filter field there is a row of buttons: *Geometry, Shapes, Lights, Cameras, Helpers, Space warps, Bones* and *Hidden, Frozen, Selected*. Category chips combine with “or” (an object has one category), state chips with “and”. They are **prepended** to the text in the field, so *Lights + Hidden* + a typed `layer:02_*` means: hidden lights from those layers. The category chips show the number of objects in the scene. The `x` button clears every chip and leaves the text. The bar can be switched off in the preferences or from the palette.

MEASURED

Filtering by class and layer compares indices, not text: **0.05–0.08 ms** on a scene of 50 026 objects. The worst case — a text pattern over the names — **4.35 ms**, against a frame budget of 16 ms.

04 Columns

Columns are chosen and reordered in *Columns...* on the panel bar. The choice is stored in the preferences.

COLUMN	WHAT IT SHOWS
Name	The only one that cannot be switched off — it holds the hierarchy indent, the class icon and the expand triangle.
Class	The class name, not localised.
Layer	The layer the object belongs to.
State	An eye and a snowflake. Clickable — a click toggles that one object, regardless of the selection.
Colour	The wireframe colour as a swatch and a hexadecimal value.
Material	The material name. Empty = no material.
Faces	The face count, computed lazily . Three dots mean “not counted yet”, never zero.
Instances	×12 — twelve instances. R3 — three objects on a shared base object.
Mod.	The number of modifiers in the whole chain. An empty field = zero, three dots = not counted yet.
Depth	How deep in the hierarchy the object sits.
Children	The number of direct children of the node. Sortable.
Verts	The vertex count from the top of the stack, noted while counting faces — it costs no separate evaluation. Three dots = not counted yet, a dash = not geometry.
Size	The largest bounding box dimension in world space, in scene units; the same one used by size: . Computed lazily for the visible rows.
Flags	State letters: H hidden, F frozen, S selected, G group head, g group member. The same as the icons in the State column, but as text — it copies into CSV.
Path	The ancestors from the root, separated by slashes. Useful in flat mode, where the hierarchy indent disappears.

The Children, Verts, Size and Path columns travel through a separate core channel and are fetched **only when one of them is switched on** — a column nobody turned on costs nothing.

ABOUT THE FACE COUNT

`getPolygonCount` reads **the top of the modifier stack**, not the base mesh.

Weighted Normals can turn 102 152 vertices into 400 824 — the Faces column then shows the effect of the stack, and the Mod. column its cause.

05 Lenses

Instead of two separate windows (Hierarchy and Layer) there is one view and a grouping switch.

LENS	ARRANGES THE SCENE
no grouping	a plain parent–child hierarchy
by layer	the equivalent of the Layer Explorer
by class	a bucket per object class
by material	a bucket per material
by material (tree)	Multi/Sub-Object expanded into its submaterials
by instance	one row per mesh, the objects inside it
by group	a bucket per nearest group head (nested groups count the nearest one), everything else in (<i>outside a group</i>)
by face count	buckets by order of magnitude: below 1 000, 1 000–9 999, ... The first use counts the faces of every object (the same cost as <code>poly: </code>), after that it comes from memory

VIEWPORT COLOURING

From the palette: *Viewport: colour objects by layer / class / material / face count*. The wireframe colour of every object is replaced (faces: a green → red gradient by order of magnitude — a heat map of the scene's weight). **This is a change to the scene:** a violet bar above the list is there to remind you, and *Restore wireframe colours* brings back the originals (also when the panel is closed, and through Ctrl+Z).

SCENE RULES

Scene rules... from the palette: a filter query + an action (hide, freeze, select, as a box, See-Through, renderability, move to a layer). A rule acts on **the whole scene**, including objects the filter is not showing. *Test* says how many objects match before you change anything; *Run all* runs the list one by one. The rules live in a file next to the preferences.

NESTING LAYERS

In the layer list **drag a layer onto another layer** to nest it underneath; drop it on an empty part of the list to pull it back to the root. Max itself refuses a cycle (a parent under its own child) — the status bar then says how many layers were rejected. The operation can be undone.

DRAGGING A MATERIAL

In the *by material* lens a bucket header can be **dragged onto an object row**. Dropping it on a row that belongs to the selection assigns the material to the whole selection; on a row outside it — only to that row. The (*no material*) header dropped onto an object takes its material off. The identity of a material is the object in the bucket, not the name — two materials can share a name. The operation can be undone.

06 Shortcuts

They work while the panel has focus. Max's own shortcuts are not intercepted — **measured**: control outside the panel 4/4, focus in the list 4/4, `Ctrl+Z` 3/3, and in the filter field the shortcuts are correctly disabled so that the letter reaches the field.

KEY	DOES
F2	rename in place
Ctrl+F2	batch rename, with a preview
Ctrl+P	the command palette — objects, actions and layers at once
Ctrl+A	select everything visible
Ctrl+I	invert the selection

KEY	DOES
Ctrl+D	clear the selection
Alt+Enter	object properties
Ctrl+Alt+Q	isolation — in and out
Ctrl+Alt+F	the filter drives visibility in the viewport
Ctrl+Alt+S	“selected only” mode
Ctrl+Alt+K	choice of visible columns
Ctrl+Shift+R	scene reports
Ctrl+Shift+L	layer states
Ctrl+1...9	saved views

Your own assignments: *Preferences* → *Shortcuts*. They travel the same path as the built-in ones, so there are not two classes of shortcut behaving differently.

07 Scene reports

Ctrl+Shift+R. Four tabs, none of which computes itself when the window opens — scanning a large scene takes time, and a window that freezes Max on opening gets opened once.

Geometry duplicates

Merge ALL walks every group in the table, checks the vertices and merges them into instances — 340 identical chairs become one mesh and 340 instances. A group with different geometry is skipped and named in the summary. Each group is its own undo block.

Groups of objects with an identical signature, sorted descending. Merging into instances **first compares the vertex positions** and refuses when they differ — two meshes with the same face count and the same dimensions need not be the same mesh.

2849 ms

scan of a whole 50 026 object scene

1.47 ms

group members from memory

Scene weight

Faces and objects per layer and per class. The face count is lazy in the snapshot, so the report computes it first — and **says how many remain unknown**. Counting them as zero would give totals smaller than the truth and looking perfectly reasonable.

Textures

Every bitmap map: what weighs how much, and what is missing. **One row per file, not per occurrence** — the same map used a hundred times does not take up a hundred places on disk. The size is the file size, not the memory footprint: that depends on compression and mipmaps and cannot be derived from the resolution. A file that is not there has the size `?`, not zero.

Snapshot comparison

Remember the baseline stores the state of the whole scene by handle (name, layer, material, class, parent, faces). *Compare* shows what was added, removed and changed — one difference per row, with CSV export. *Show the changed ones in the list* writes `is:changed` into the filter. An audit of “what have I touched since this morning” in a single table.

Timeline

The most recent scene events (added, deleted, structure, name, state) with a time and a name — a ring of 5 000 events. The *in the list* writes `changed:<N m` into the filter.

Scene card

A single HTML file with the numbers about the scene: objects, layers, faces, the heaviest layers and classes, scene health, textures, duplicates. To send to somebody who does not have Max. *Compute* gathers the reports (the duplicate scan takes about 2 s on 50 000 objects), *Save HTML...* opens the file in a browser.

Naming

Rules in the form `name|regular expression`, one per line. An object breaks the convention when it matches **none of them**. A bad pattern shows `ERROR` instead of pretending that nothing matches — a typo in a pattern would report the whole scene as a deviation, and that would look like a genuine result.

The names found can be fixed in the same window. The fix works by **handles**, not by view rows: a filtered-out object has no row number, so going through the view would fix 40 names out of 300 and report “40 changed”.

UNDO

The whole fix is one `Ctrl+Z`. On top of that, a CSV file with the old names is written next to the scene before the change — the undo history does not survive saving and closing the scene, but that file does.

08 Explorer Turbo

A separate, free tool for the **native** Max Scene Explorer. It does not require SceneForge and does not carry its code.

- **Suspending the refresh** with a single switch, with the state visible on the bar.
- **Four columns** the native explorer does not have: modifiers, children, instances and a short note on node health.

WHAT IS NOT THERE, AND WHY

There is no face-count column. Reading it from MAXScript costs **311 µs per object**, that is 15.6 ms per 50 visible rows — nearly a whole frame. And that on light primitives, so for real geometry it is a lower bound. A tool meant to cure a sluggish explorer cannot be what makes it sluggish.

What does go in costs, all together, **0.856 ms per 50 visible rows**. One thing saves us: Max only asks about the rows you can see — measured at about 50 on opening and zero growth after two seconds of idling.

09 Performance

The same operations, the same scene, the same Max session. The minimum of a series, not the average.

OPERATION	NATIVE EXPLORER	SCENEFORGE	RATIO
opening and filling	92 992 ms	101 ms	920×
expanding the whole tree	3 383 298 ms	1 ms	>10 ⁶ ×
cost of an idle window	+0 %	+0,9 %	a draw
update after adding 1000 objects	—	4.4–6.3 ms	see below

Three things in this table that are easy to read wrong:

- **An idle native explorer costs nothing.** The common form of the complaint — “an open explorer slows Max down” — is untrue as stated. The cost sits in filling and rebuilding the list.
- **The last row has no “native” column, and that is the honest answer.** Creating 1000 objects costs Max itself 8.7–12.9 s, and the spread between identical runs reaches 47 %. An overhead of a few milliseconds cannot be seen inside that — on either side. So we quote what was measured directly: our own snapshot update.
- **Our “1 ms” is real work.** The view grows from 33 812 to 50 026 rows and hands back data from the start, the middle and the end of the list. It is cheap because the view is an array of indices, and only the rows you can see are drawn.

Busy signal

Operations on a large selection can take seconds. From 500 objects up, the panel says in the status bar what it is doing and switches the cursor to the busy one. The threshold was not picked by eye: at 500 objects the slowest of those operations takes 67 ms, which is imperceptible, while at 1000 it is already 214 ms. The signal itself costs 3.4 ms and appears only where you are waiting anyway.

A CORRECTION

An earlier version of this page quoted “+111 ms for us against +6 ms for the native one” here, and the sentence “this is where we are slower”. Checking that

charge demolished **the measurement itself**, not the code: both numbers were samples from noise 37 times larger than they were. The method “measure the whole operation and subtract” had no chance of seeing an effect of a few milliseconds.

10 Footprint in the scene

A `.max` file is a compound document with eleven streams. After a full session of use — refreshing, filters, lenses, face counting, every report, a duplicate scan — a stream by stream comparison:

STREAM	WITHOUT THE PLUGIN	AFTER A SESSION
Scene — the scene data, 287 078 B	daf529c31a270f0c	daf529c31a270f0c
Config	68d949cbd1c3fc17	68d949cbd1c3fc17
the remaining eight	identical	
SummaryInformation — the viewport thumbnail	differs between two saves as well without the plugin	

Not one word of ours — `SceneForge` , `SF_CORE` , the column names — appears in the saved file.

ONE EXCEPTION, A DELIBERATE ONE

You save layer states yourself, with the command `Ctrl+Shift+L` . They then go into the `Config` stream, because a layer state describes *this* scene and means nothing in another one. It is the only thing the plugin puts into the file — and only when you ask for it.

11 Troubleshooting

SYMPTOM	WHAT TO DO
I want to send a log to support	The panel writes its log to <code>dziennik_panelu.txt</code> in the configuration folder (next to <code>preferencje.json</code>); the scene change listener writes to <code>SceneForge_events_log.txt</code> in the Max user scripts folder.
The list is empty although the scene is not	The core and the panel have drifted apart in version. Reload with the command from Customize; if that does not help, install the package again — the installer removes the old versions.
The Faces column shows three dots	That is how it should be. The face count is computed for the rows you can see, with a budget of 2 ms per pass. Scroll, or wait a moment.
There is no SceneForge entry in the menu	It appears after 3ds Max is restarted. The commands in Customize work right away.
The naming report flags the whole scene	The default rules (<code>SM_</code> , <code>HLP_</code> , <code>LGT_</code> , <code>CAM_</code>) are examples meant to be replaced. Type your own and save.
Merging duplicates refuses	The meshes have the same face count and the same dimensions, but different vertex positions . The refusal is the correct result here.
I cannot see what the object got parented to	The parent is filtered out. Clear the filter, or switch on auto-expand to the selection in the preferences.

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